AI Hallucination Detector

A major project report submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

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Supervisor's Certificate

This is to certify that the major project report entitled **'AI Hallucination Detector'**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is a bonafide project work carried out under my supervision during the period from July 2025 to May 2026.

I have personally supervised the research work and confirm that it meets the standards required for submission. The project work has been conducted in accordance with ethical guidelines, and the matter embodied in the report has not been submitted elsewhere for the award of any other degree or diploma.

Supervisor Name: Prof. Dr. Vivek Sehgal

Date Designation: Professor & HoD (CSE & IT)

Place: Department: Computer Science & Engineering

Candidate's Declaration

We hereby declare that the work presented in this major project report entitled **'AI Hallucination Detector'**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is an authentic record of our own work carried out during the period from July 2025 to December 2025 under the supervision of **Prof. Dr. Vivek Sehgal.** We further declare that the matter embodied in this report has not been submitted for the award of any other degree or diploma at any other university or institution.

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LIST OF ABBREVIATIONS, NOMENCLATURE

Abbreviation/Symbol	Full Form
AI	Artificial Intelligence
LLM	Large Language Model
SRL	Semantic Role Labeling
RAG	Retrieval- Augmented Generation
BLEU	Bilingual Evaluation Understudy
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
ACL	Association for Computational Linguistics
GPT	Generative Pre-trained Transformer
NLP	Natural Language Processing
QA	Question Answering
PDF	Probability Density Function
GAT	Graph Attention Network

ABSTRACT

Artificial intelligence system which mainly includes various multimodal recently got a lot of attention because of their outputs , the reasoning that they give , the analysis that they made and so on. But we haven't gone through one thing in this which is acting as a very big problem which affects the output and affects the trust of all of us which is nothing but hallucinations. But why it is happening? That's the main question to all of us. What exactly is wrong in all AI Models that they are hallucinating. Think about it. It's a very big problem in sectors such as health,law and finance especially, one term changes the whole situation, so correct information is really important.

Why have we taken this project? To understand why this is happening, what is the root cause of all this , how to detect the hallucinations and how we can reduce it. Though their are many models and applications that are serving this thing but we have in our mind that how can we apply some unusual things on the model. So we are trying to apply a model which can automatically check things , applying attention based features, taxonomy based evaluation not only for text but also for videos images and audios.

So what we have seen till now that the datasets that models are using is still facing some kind of issues which lacks reasoning, proper understanding and a lot of instructions . We have seen that the ways that are used to reduce hallucinations till now helps us but not completely and not on a satisfactory level. So to make models reliable we have to apply a lot of changes to improve the reliability and accessibility to the understanding, reasoning, set of instructions, datasets etc.

By this project we are looking firstly into the problems that comes in the datasets like data leakage, cost problems and various other problems. We need to make a model which helps human to trust Artificial intelligence much more and to become trustworthy so that everyone can apply AI in real world applications specially in the areas of law finance and healthcare.

Chapter 1: INTRODUCTION

1.1 Introduction

Artificial intelligence system which mainly includes large language models and various other models have gotten popular so much because of the solutions that they are genrating , the reasoning that they are giving , the analysis that they are making and so on.

But all these abilities are not making full impact on all the real world applications because of one serious problem of hallucination.

What is Hallucination? When AI generates an answer which looks very convincing to us but it is not actually true.Main reason for that is that they generate responses on the basis of the pattern that they are trained on , not the actual facts.

- That clearly means that if they are not completely aware about the actual facts they tend to make mistakes.
- If we try to understand real world applications, hallucination is a big problem in sectors like:
    - Medical/ Health – Probability of getting wrong answer.
    - Law – False Allegations.
    - Finance – Wrong Transaction Statements

Artificial Intelligence has transformed the way humans interact with technology. From virtual assistants and recommendation systems to medical diagnosis and autonomous systems, AI has become an important part of modern society. Among all AI technologies, Large Language Models (LLMs) such as GPT, BERT, Gemini, and LLaMA have gained significant popularity because of their ability to generate human-like responses and perform reasoning-based tasks.

These models are trained on massive datasets collected from books, websites, research papers, and online conversations. Due to such large-scale training, they can generate highly fluent and context-aware responses. However, despite their impressive capabilities, these systems still face one major challenge known as hallucination.

AI hallucination refers to the generation of information that appears convincing but is actually false, misleading, or unsupported by facts. In many cases, the generated content sounds grammatically perfect and logically structured, making it difficult for users to immediately identify errors. This becomes dangerous in domains where factual accuracy is critical.

For example, in healthcare systems, an AI model may provide an incorrect medical recommendation or suggest a non-existent treatment. In legal systems, hallucinated references or fake case citations can create serious consequences. Similarly, in finance and education, incorrect AI-generated information may lead to poor decisions and misinformation.

One of the main reasons behind hallucinations is that language models are fundamentally prediction systems. Instead of truly understanding facts like humans, they predict the next probable word based on learned patterns. When the model lacks sufficient knowledge or context, it may generate fabricated information with high confidence.

Another important factor is the quality of training data. If the training dataset contains incomplete, biased, or outdated information, the model may learn incorrect associations. Additionally, decoding techniques such as temperature sampling and beam search also influence hallucination frequency.

As AI systems continue to become part of real-world applications, reducing hallucinations has become a major research challenge. Organizations and researchers are now focusing on building systems that are not only intelligent but also trustworthy, explainable, and reliable.

The proposed project, AI Hallucination Detector, aims to address this challenge by creating a framework capable of identifying factually incorrect outputs generated by AI systems. Unlike traditional evaluation methods that mainly compare textual similarity, this project focuses on semantic consistency, factual correctness, and graph-based relational learning.

The system combines Natural Language Processing techniques, semantic embeddings, and Graph Attention Networks (GATs) to analyze relationships among generated responses.

By learning semantic structures and consensus patterns, the system can identify hallucinated content more effectively than traditional standalone methods.

The long-term vision of this project is to contribute toward safer and more dependable AI systems that can be confidently used in sensitive real-world applications.

1.2 Problem Statement

While LLMs are very good at generating natural and human-like responses their habit of creating hallucinated content makes them less reliable and less trustworthy.

- Current evaluation metrics like BLEU and ROUGE mainly check how similar the generated text is to the reference text in terms of language. They don't check if the information is actually correct, which means a text can get a high score even if it is factually wrong.

- The existing methods to reduce hallucination are also not good enough because:

  - Many of them work only for a specific field (for example, they might work in medicine but fail in other areas).
  - Some of them need too much computing power so they can't be used in real-time applications.
  - Others are not clear enough they give a score but don't explain why something is wrong.

So, there is a clear need for a hallucination detection framework that is:

- Effective → can correctly find factual mistakes.

- Interpretable → can explain the reason, not just give a number.

If system like we are discussing comes in picture , it will be a boom for AI Industry and it will be really beneficial to real World.

Although modern Large Language Models are capable of generating highly sophisticated responses, they still suffer from the issue of factual inconsistency. Existing models often generate statements that are grammatically correct and semantically fluent but factually inaccurate. This creates a major reliability problem for practical deployment.

Traditional evaluation metrics such as BLEU, ROUGE, and METEOR mainly focus on textual similarity between generated content and reference text. These metrics fail to evaluate whether the information itself is factually correct. As a result, a response containing false information may still receive a high evaluation score.

Current hallucination mitigation techniques also have several limitations. Retrieval-Augmented Generation systems require continuous access to external knowledge databases, which increases computational complexity. Similarly, methods like SelfCheckGPT generate multiple responses to verify consistency, making them computationally expensive for real-time systems.

Another challenge is the lack of explainability in existing detection methods. Many systems only provide a hallucination score without explaining which part of the response is incorrect or why the model considered it hallucinated. This lack of transparency reduces user trust.

Furthermore, most existing research focuses only on English-language text-based systems, while multilingual and multimodal hallucinations remain underexplored.

Therefore, there is a strong need for a hallucination detection framework that is:

- Accurate in identifying factual inconsistencies
- Computationally efficient
- Explainable and interpretable
- Adaptable across multiple domains
- Scalable for real-world applications

The proposed system addresses these challenges by combining graph-based semantic analysis with attention-based learning techniques.

1.3 Objectives

Let's discuss what are the main objectives of this project:

1. **Main objection is to understand why hallucination occures in Large Language Models(LLMs).**

Conversation should include what is hallucinations and why it occurs in various models.Then it should include how hallucinations occur and how it checks the tokens and

Pattern of the data that is provided to them, so let us see where things can go wrong in the dataset:

- Unclear or confusing user questions.
- Rare or new situations that the model has never seen before.
- The model being too confident even when it is unsure.

By studying these reasons, the project will create a strong base for detecting and reducing hallucinations in a better way.

2. **To check the detection andmethods through a detailed study.**

First we will start by reading and researching all the research papers on this topic to understand the problem and its solution well.

This includes checking different methods like:

○ Detection approaches such as black-box scorers white-box scorers, retrieval augmented generation (RAG) and LLM-as-a-judge techniques. Strategies are using various models to check the coding and encoding schemas of various datasets.

This review will help us to determine the gaps between various models , various datasets and it will help us to understand that where we have to pick up to start for the project.

This step will make sure that the project builds on the latest research instead of repeating what has already been done.

3. To create and propose an AI hallucination detection framework that checks factual correctness in generated text.

Based on what is learned from the research, the project will design a new detection framework that has a modular structure.

The main parts of this framework will include:

- **Fact Extraction Module:** Changes sentences into fact-based structures.

- **Scoring Modules:** Employs various approaches including black box,white box,LLM-as-a-judge,and combined (ensemble) scoring schemes.

- **Hallucination Detection Module:** Combines the scores of everything and makes a decision to label a false or unsupported statement

- **Output Module:** Brings simple transparent easy-to-read reports with the sections of text containing hallucination.

- **To provide ideas to make AI more reliable and trustworthy**

○ In the end, the project will provide beneficial recommendations and best practices from research community, developers and people working in AI industry.

○ The main moto is to provide a better future AI work so that it becomes safer, better and trustworthy.

1.4 Significance And Motivation OfThe Project Work

The whole moto that comes under the growing of Artificial Intelligence(AI) is that in this accuracy should beat the top notch.

We can see the importance by the points given:

- **Trustworthiness:** Give more reliability to the models by reducing false or made-up responses.
- **Practical Impact:** Helping to use LLMs safely in important areas like healthcare, law, finance,and education.
- **Contribution of the Research**: Provide a proper framework for finding hallucinations that can serve as the basis for further work toward making AI safer**.**

The development of this project is not only important from a research point of view but also has great social impact since it focuses on solving one of the biggest problems in today's AI usage.

1.5 Organization of Project Report

The six chapters of this project report are systematically organized to present the complete development and evaluation of the proposed AI Hallucination Detection System. Each chapter focuses on a different stage of the project, beginning with the background and motivation behind the work and progressing toward implementation, experimentation, and final conclusions. The arrangement of chapters allows readers to clearly understand the logical flow of the project from identifying the problem of hallucination in Artificial Intelligence systems to proposing and evaluating a graph-based detection framework.

Chapter One: Introduction

This chapter introduces Artificial Intelligence and Large Language Models (LLMs) and explains the problem of AI hallucination, where systems generate incorrect but convincing information. It discusses the project objectives, motivation, significance, and the need for reliable hallucination detection systems.

Chapter Two: Literature Survey

This chapter reviews existing research on AI hallucination detection techniques, including RAG, uncertainty estimation, graph-based learning, and multimodal analysis. It also identifies research gaps such as limited explainability, high complexity, and lack of real-time support.

Chapter Three: System Development

This chapter explains the design and implementation of the proposed AI Hallucination Detection System. It describes dataset preparation, semantic embeddings, graph construction, Graph Attention Networks (GAT), training methods, and development tools used in the system.

Chapter Four: Testing

This chapter presents the testing methods and evaluation process of the proposed system. It discusses dataset splitting, performance metrics such as accuracy and F1-score, and compares the proposed model with baseline approaches.

Chapter Five: Results and Evaluation

This chapter presents the experimental results and performance analysis of the system using visualization techniques and comparative evaluations. It explains how graph-based semantic learning improves hallucination detection performance.

Chapter Six: Conclusion and Future Scope

This chapter summarizes the project contributions and highlights the effectiveness of graph-based semantic analysis for hallucination detection. It also discusses future improvements such as multilingual support, explainable AI, multimodal analysis, and real-time optimization.

CHAPTER 2: LITERATURE SURVEY

2.1 Overview of Relevant Literature

AI and NLP have advanced very rapidly throughout the last decade-widely-general top performances have been achieved with techniques like large language models (GPT BERT LLaMA, ..). Yet, one tend to get recurrently-frequent area of concern in these successful projects:problem of hallucination, in which LLMs produce a text which seemsto be correct and lucid but might be genuinely "mistaken". Here is an overview of recent work done on hallucination from 2019- 2024and how to detect, mitigate the hallucination. Various types of hallucinations and their definitions have been attempted to be specified and understood in several research works.

Huang et al.[15] proposed a classification scheme for hallucinations in LLMs by categorising them into two broad groups: factuality hallucinations, contrary to facts of the real world and faithfulness hallucinations, which could be incongruent with the user prompt or the network's logic. Their work also categorized the instances in which hallucinations may arise. during data, training and inference by also referring to several evaluation benchmarks like TruthfulQAand HaluEval.

Zhang et al. [14] in their "Siren's Song in the AI Ocean" survey emphasized that it's not just about the bad quality of the data, it's also about how the model produces the hallucinations, for instance, text through decoding techniques like a temperature sampling.

They further elaborated about art versus factuality and proposed useful methods like Retrieval- Augmented Generation (RAG), consistency checking, and uncertainty estimation for hallucination reduction. Other important contributions include:

TruthfulQA introduced by Lin et al. [18] is an evaluation benchmark for the capacity of language models to produce false but plausible information.

SelfCheckGPT (Manakul et al.[17]) checks for hallucination by generating different outputs from the same model and verifying the consistency between the two outputs.

REALTIMEQA (Kasai et al. 2007) uses Kasai et al. to asses hallucinations produced in questions answering tasks where the factual knowledge is changing with time.

Hallucination in Multimodal Models (Bai et al. 2024): extended hallucination to vision-language models, in which error can occur in both image modeling and text generation.

All of these research areas combine to provide a complete understanding of hallucinations what they are, why they matter, and how ways in which researchers are working toward methods of reducing and perceptually-sensitive means of detection to create more reliable and trustworthy AI systems.

Table 1: Literature Survey

S.No	Author & Paper Title	Journal/ Conference	Tools/ Techniques/ Dataset	Key Findings/ Results	Limitations/ Gaps Identified
1	Brahmaleen Kaur Sidhu, "Hallucinations in Artificial Intelligence: Origins, Detection, and Mitigation" [24]	IJSR (2025)	Review; SelfCheckGPT, ChatProtect, GPTScore, RHO, NPH RAG, MixCL, HERMAN; datasets include WikiBio, OpenDialKG	Categorizes AI hallucinations, surveys detection/mitigation models, urges multi-method approaches	Detection tools lack context focus; output quality gaps; factual alignment remains challenging
2	Zhiyang Chen et al., "Mitigating Hallucination in Visual Language Models with Visual Supervision" [25]	arXiv (2023)	Visual annotation (PSG, SAM), Mask prediction, RAH-Bench	Fine-grained annotation, mask prediction, and visual supervision help LVLMs reduce three types of hallucination and outperform baseline models	Data format coupling; limited dataset size; method needs generalization across annotation formats
3	Dylan Bouchard et al., "UQLM: Uncertainty Quantification for Language Models" [26]	arXiv (2025)	Python package uqlm; UQ-based scoring methods (Black-Box, White-Box, LLM-as-Judge), multiple existing benchmarks	Provides library for uncertainty-based hallucination detection at generation time; democratizes advanced quantification for LLM outputs	Limited technique choice in prior tools; adoption and ease-of-use for non-experts remains; external fact-checking still a challenge
4	Adam Tauman	Preprint	Theoretical	Hallucinations arise	Benchmarks

	Kalai et al., "Why Language Models Hallucinate" [27]	(2025)	analysis, meta-evaluation of LLM benchmarks	from statistical pressure in training; binary evaluations reward guessing over uncertainty	penalize uncertainty; socio-technical fixes needed in evaluation/grading; open questions on uncertainty signaling
5	Zhang et al., "Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models" [14]	arXiv preprint (2023)	Literature review across detection & mitigation strategies	Showed hallucinations are systemic; reviewed detection (fact-checking, NLI, uncertainty) and mitigation (RAG, consistency, instruction tuning)	Conceptual; lacks empirical evaluation; limited to text-based LLMs
6	Chaoyou Fu et al., "MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models" [5]	arXiv Preprint (2024)	Technique: MME Benchmark with manual yes/no instruction-answer pairs; Models: 30 MLLMs (e.g., GPT-4V, LLaVA); Metrics: Accuracy, Accuracy+	Benchmarked perception (existence, position, OCR, etc.) and cognition (reasoning, calculation, translation, code) abilities for 30 models. Identified persistent gaps: failure to follow instructions, perception errors, object hallucination, and reasoning limitations	Models struggle with concise instructions, object perception and universal robust reasoning. Data leakage risk and challenge of instruction generalization. Not all cognitive tasks achieved high accuracy; further benchmarking needed
7	Xu et al., "Hallucination is Inevitable" [8]	arXiv 2024	Theoretical proof using learning theory & diagonalization. Defines a formal world with computable LLMs	Proves hallucination is mathematically inevitable for any computable LLM, regardless of architecture or training data. Identifies hallucination-prone problem classes	Analysis is theoretical; does not quantify real-world hallucination rates. Assumes deterministic ground truth, limiting probabilistic insights. Focuses on inherent limits
8	Maleki et al., "AI Hallucinations: A Misnomer Worth Clarifying" [9]	Preprint (e.g., arXiv) 2024	Systematic review of 14 databases from 2013–2023. Manually extracted and	Highlights inconsistent and often contradictory definitions across domains. Suggests alternative terms	Manual review may miss newer definitions. Focuses on terminology, not technical solutions.

			analyzed 333 definitions of "AI hallucination"	(e.g., fabrication, stochastic parroting). Calls for more precision	Limited to English-language papers. Does not propose a unified taxonomy
9	Pranab Sahoo et al., "A Comprehensive Survey of Hallucination in Large Language, Image, Video and Audio Foundation Models" [7]	arXiv Preprint (2024)	Technique: Systematic taxonomy and review of detection/mitigation strategies	Provides a unified framework and synthesizes detection/mitigation strategies	Scope is broad, not deep. Cuts off at May 2024, missing recent works
10	Chaoyou Fu et al., "MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models" [Fu et al., arXiv:2306.13394]	arXiv Preprint (2024)	Technique: MME benchmark with manual yes/no instruction-answer pairs; Models: 30 MLLMs (e.g., GPT-4V, LLaVA); Metrics: Accuracy, Accuracy+	Benchmarked perception (existence, count, color, position, OCR, etc.) and cognition (reasoning, calculation, translation, code) abilities for 30 models. Identified persistent gaps: failure to follow instructions, perception errors, object hallucination, and reasoning limitations	Models struggle with concise instructions, object perception, and universal robust reasoning. Data leakage risk and challenge of instruction generalization. Not all cognitive tasks achieved high accuracy; further benchmarking needed
11	Cheng Niu et al., “RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models” [28]	arXiv Preprint 2024	Used GPT-3.5, GPT-4, Llama-2, Mistral-7B to generate responses; Detection methods: Prompt-based (GPT), SelfCheckGPT, LMvLM; Fine-tuned Llama-2-13B; Dataset: RAGTruth	Data-to-text had the most hallucinations (68%); GPT-4 the least; Mistral-7B the most. Fine-tuned Llama-2-13B reached 78.7% F1 and cut hallucinations by 50–63%	Focused only on RAG and three tasks; span-level detection weak; annotations subjective; results may not generalize. Need bigger diverse datasets, better subtle hallucination handling, cross-task/multilingual tests
12	Junyang Wang et al., “AMBER: An LLM-free Multi-	arXiv Preprint 2024	Proposed AMBER, an LLM-free benchmark for hallucination evaluation in	GPT-4V performed best with the least hallucinations and highest AMBER score, followed by	AMBER only evaluates attribute and relation hallucinations in discriminative

	dimensional Benchmark for MLLMs Hallucination Evaluation” [29]		Multi-modal LLMs (MLLMs); Dataset: MSCOCO Test set	Qwen-VL and InstructBLIP. Object models still showed hallucinations, especially in attribute and relation cases	tasks, not generative ones. Object extraction errors may occur
13	Balaji Padmanabhan et al., “AI Hallucinations : A Misnomer Worth Clarifying”	arXiv 2024	Systematic literature review (14 databases, 333 papers), manual screening, categorization by domain	No consistent definition of “AI hallucination”; term used inconsistently; alternatives like confabulation, fabrication, misinformation	Focus on definitions not solutions. Need unified taxonomy, standardized non-stigmatizing terms
14	Varun Magesh et al., “Hallucination -Free? Assessing the Reliability of Leading AI Legal Research Tools” [30]	Journal of Empirical Legal Studies (2025)	Empirical tests on Lexis+ AI, Westlaw, Practical Law AI, GPT-4 with 200+ legal queries, hand-coded evaluation	All still hallucinate (Lexis+ ~17%, Westlaw ~33%, Practical Law AI high incompleteness worst); GPT-4 worst; RAG helps but not a fix	Proprietary black boxes, small dataset, US-law only, snapshot in time. Need bigger cross-jurisdiction studies, better benchmarks, lawyer-side mitigation, long-term ethical analysis
15	Fan, Aumiller, and Gertz, “Evaluating Factual Consistency of Texts with Semantic Role Labeling” [16]	ACL 2023 Proceedin gs	Semantic Role Labeling (SRL); datasets: QAGS (CNN/XSUM), SummEval	Proposed SRLScore (reference-free factuality metric); achieved high correlation with human judgment; interpretable	Slower than LM-based methods; only tested in English; occasional mismatch with human evaluations
16	Guliyev and Özer, “On the Hallucinations of Artificial Intelligence and Their Terminologies ” [31]	Cureus (2024)	Conceptual analysis of terminology	Argues “hallucination” is metaphorical, not literal; proposes using terms like fabrication/confabul ation instead	Theoretical only; no technical solutions
17	Huang et al., “A Survey on Hallucination in Large Language Models” [15]	ACM Computing Surveys (2024)	Literature survey of LLM hallucination studies	Provided taxonomy of hallucinations (factual vs. faithfulness); identified causes (data, model, decoding); reviewed benchmarks like	Survey only; no new method proposed; limited multimodal/multili ngual coverage

				TruthfulQA and HaluEval	
18	Santosh S. Vempala et al.	arXiv Preprint 2025	Computational learning theory, binary classification, reduction, error analysis, benchmark evaluation review, confidence-targeted evaluation	Hallucinations are statistically inevitable; caused by arbitrary facts, poor models, and evaluation design; benchmarks reward guessing over "I don't know."	Mostly theoretical, simplified binary framework, focused on factual QA, little empirical validation. Need fine-grained taxonomy, large-scale empirical tests, domain-specific study, and better evaluation metrics
19	Jeffrey P. Bigham et al.	arXiv preprint / Presented at Human-Centered Machine Learning Workshop (2024); Scheduled at ACL (Vienna, 2025)	MMAU Benchmark; PLUM (Pipeline for Learning User Conversations in LLMs); External Validation Tools	MMAU provides more comprehensive and interpretable evaluation compared to older benchmarks. PLUM improves personalization by leveraging conversational history beyond static user preferences. External validation sometimes improves annotation quality and reduces biases in LLM-as-a-Judge setups	External validation tools improved results only "often, but not always". Personalization in PLUM still limited to extracted Q-A pairs. Apple strong in research output, but practical deployment into consumer-facing AI products lags behind competitors
20	Alkaissi and McFarlane, "Artificial Hallucinations in ChatGPT: Implications in Scientific Writing" [20]	Cureus Journal of Medical Science (2023)	Tested ChatGPT on medical topics (Homocystinuria, Pompe disease)	Showed ChatGPT fabricated citations and unsupported claims; highlights risks in medical/scientific domains	Case-based only; not systematic; no general detection method proposed

Recent advancements in Artificial Intelligence have significantly improved the capabilities of Large Language Models. However, as these systems become more powerful, the issue of hallucination has also become more noticeable. Researchers across different domains have proposed several methods for understanding, evaluating, and reducing hallucinations.

One of the major research directions focuses on factual consistency evaluation. Researchers have attempted to compare generated outputs with trusted reference documents in order to identify factual mismatches. Semantic similarity methods, Natural Language Inference (NLI), and question-answering-based evaluation approaches have gained popularity in this area.

Another important research area is uncertainty estimation. Several researchers argue that hallucinations occur because models generate highly confident outputs even when they are uncertain. Methods such as entropy-based scoring and token probability analysis have been introduced to estimate model confidence during generation.

Retrieval-Augmented Generation (RAG) has also emerged as a powerful hallucination reduction strategy. In RAG systems, external knowledge databases are connected to language models so that the model can retrieve factual information before generating responses. While RAG improves factual grounding, it also increases computational complexity and dependency on external retrieval systems.

Graph-based learning methods are another emerging direction in hallucination research. Unlike traditional approaches that evaluate responses independently, graph-based methods study relationships between multiple generated responses. Similar responses form clusters, while hallucinated responses often appear isolated or weakly connected. This idea inspired the graph-based architecture used in the proposed project.

Researchers have also explored hallucination detection in multimodal systems involving text, images, and videos. In visual-language models, hallucinations may occur when the model describes objects that are not present in an image or misinterprets visual information. This demonstrates that hallucination is not limited to textual systems alone.

Despite these advancements, several challenges remain unresolved. Many proposed methods are computationally expensive, domain-specific, or difficult to interpret. Additionally, the lack of standardized benchmarks makes fair comparison difficult.

The literature clearly indicates that hallucination detection requires a combination of semantic understanding, factual verification, uncertainty estimation, and explainability. Therefore, hybrid approaches integrating multiple techniques are likely to provide better performance.

2.2 Key Gaps In The Literature

Lack of Benchmarks

Currently, the standards used to spot hallucinations are very limited and rely on specific tasks For example:

- In healthcare, datasets focus on checking if the model gives factually correct diagnoses.
- In news, datasets check whether the summaries match the original articles.

But there is no single benchmark that works across different areas like health, law, education, finance, general Q&A, or creative writing. In simple words we can say that, without a common & standardized test it can creates difficulty to compare different methods with a proper systematic way .

1. Overreliance on Model-Based Metrics

Most available evaluation methods today, including BARTScore or QA-based evaluators, rely on. It allows other large models to decide if an answer is a hallucination.

- BARTScore relies on a pre-trained model to assess the closeness of a generated
- Problem: These "judge models" can have their own biases and mistakes, which makes their results less trustworthy.
- Also, they act like black boxes they can mark a sentence as hallucinated but can't clearly explain why.

This creates a circular problem: using one imperfect model in order to evaluate another one.

2. Limited Real-Time Applicability

Some methods like RAG (Retrieval-Augmented Generation) or SelfCheckGPT lot of computing power.

a. RAG constantly retrieves information from external databases.

b. SelfCheckGPT makes multiple versions of an answer to compare and check for consistency

These techniques are too heavy to be used in real-time applications such as:

a. Customer support chatbots

b. Financial trading assistants

c. Emergency healthcare bots

But, this is still a huge disconnect between the experimental results and what can practically be applied to real systems.

3. Insufficient Multilingual and Multimodal Research

a. Most studies only refer to hallucinations in the English language.

b. Very few studies have attempted to test for hallucinations in multilingual LLMs like those operating in Hindi, Spanish or Mandarin.

c. Even fewer studies have focused on multimodal models that represent text images sound or video.

For example, a multi-modal system might generate an erroneous description of the image or wrongly caption a video, but these kind of errors have not been thoroughly investigated.

4. Inadequate Handling of Uncertainity

LLMs often sound very confident, (but are completely wrong).

a. If a model is not trained on a knowledge fact. Instead of responding "I don't know", it May construct a complex but untruthful reply.

b. Only a handful of methods attempt to make models exhibits uncertainty or provide probabilistic Will find the answers.

Building LLMs that can admit when they are unsure would greatly reduce risks, especially in sensitive areas like medicine, law, and finance.

5. Trade-off between Creativity and Accuracy

a. Use algorithms to lower the randomness in text generation (like decreasing temperature or Top-k sampling).

b. How the mirror raises its 'responses' also affects the model's 'creativity' as in this solution it makes its responses repetitive, predictable and less diverse.

The things come out is balancing, its play’s major role in every area of filed special to the creative where we need brainstorming, storytelling, marketing execution,content creation.

So, researchers must find the right balance keeping outputs accurate without completely losing the model’s creativity.

CHAPTER 3: SYSTEM DEVELOPMENT

3.1 Requirements And Analysis

The development of AI Hallucination Detector has to take into account its functional requirements (what system should do) and non functional requirements (how well system should do it). Based on the project scope and what has been learned from the research the requirements are written below:

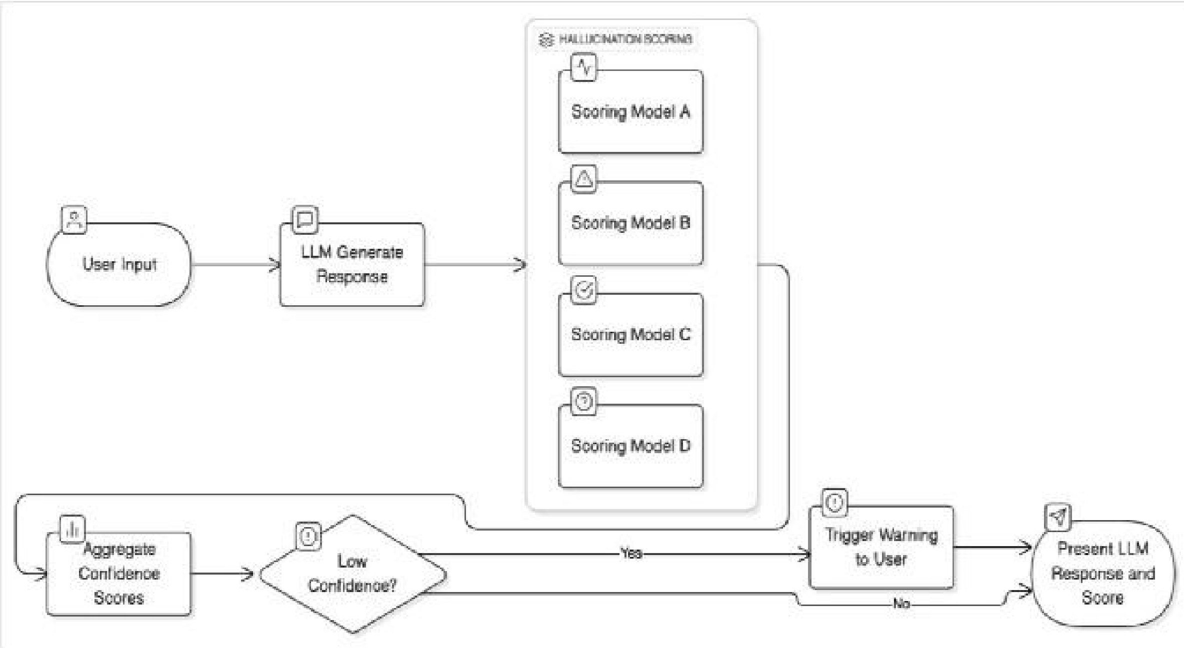

Figure 3.1 : Functional Requirements of the AI Hallucination Detector

3.1.1 Functional Requirements

The proposed structure AI Hallucination Detection System is capable of receiving and analyzing over texts formulated by large language model like GPT, BERT, and LLaMA. Currently, its input is the AI formed text responses. As an enhancement, the system also receives the reference document, or an external dataset, to perform fact verification to reference knowledge. The first component, Natural Language Processing, is activated to parse the factual information in the inputs. The extracted knowledge is then cross-checked or cross-checked with the reference knowledge to measure semantic relation and factual correctness. From the process, the model computes the result in the form of a hallucination score, estimating the likelihood of factual inconsistency. In the end, AI Hallucination Detection System generates an easy-to-understand output report with hallucinatory bits and explanation from the detection and analysis process

3.1.2 Non-Functional Requirements

Besides of the functional requirements, some other significant non-functional requirements have been considered during the designing of the proposed system. The setup should be scalable enough to cater short as well as long text responses. The structure should be interoperable to work with different large language models and external knowledge sources. Interpretable responses should be provided instead of merely scores. The system should be computationally efficient to be applicable for near real time hallucination detection applications. Adaptability of the proposed model over different domains like healthcare finance education or legal systems has also been taken into considerations. Reliability and robustness have also been favored.

3.2 Project Design And Architecture

The system proposed is designed in such a modular way,so that it is flexile and scalable.Multiple detection methods are combined together such as black-box,white-box and ensemble scorers to make the result more accurate

3.2.1 System Workflow

The proposed AI Hallucination Detection structure follows from several steps. Input module: input text generated by the LM, and optionally reference documents for fact checking. Fact extraction: use of NLP techniques (for example SRL, and dependency parsing) to discover "factual" connection between words in the input text; and passing these extracted facts to the multiple scoring modules. Scoring modules: have several scoring modules (for example, black-box, white-box, and ensemble-based). Black-box scoring modules quantify the semantic alignment between generated text and reference text, while the white-box scoring modules leverage the token probabilities and confidence measures of the LM. The scoring modules output are fed into the hallucination detection module to compute the overall hallucination score. Visualization module: visualize the detecting results on the screen (highlight reasoned segments, generate visual reports).

The workflow of the proposed AI Hallucination Detection System is divided into multiple interconnected stages. Each stage performs a specific task that contributes toward identifying factual inconsistencies in AI-generated content.

Step 1: Input Collection

The process begins with collecting AI-generated responses from Large Language Models. The system accepts user prompts along with generated outputs. In some cases, reference documents or trusted knowledge sources are also provided for verification purposes.

Step 2: Text Preprocessing

The collected responses undergo preprocessing to improve data quality. This stage includes:

- Removal of unnecessary symbols
- Lowercasing and normalization
- Tokenization
- Sentence segmentation
- Stop-word filtering

Preprocessing ensures that the system works with clean and structured text data.

Step 3: Semantic Embedding Generation

After preprocessing, each response is converted into numerical vector representations using transformer-based embedding models. These embeddings capture semantic meaning and contextual relationships between sentences.

Semantic embeddings allow the system to compare responses based on meaning rather than exact wording.

Step 4: Graph Construction

The generated embeddings are used to build a semantic similarity graph. Each response becomes a node in the graph, while edges are formed between semantically similar responses.

Highly related responses create dense clusters, whereas hallucinated or inconsistent responses often remain weakly connected.

Step 5: Graph Attention Network Processing

The constructed graph is passed through a Graph Attention Network (GAT). The GAT assigns different importance levels to neighboring nodes and learns semantic relationships between responses.

Attention mechanisms help the model focus on relevant connections while ignoring noisy or misleading information.

Step 6: Hallucination Scoring

The trained model generates hallucination scores for each response based on semantic consistency and factual alignment. Responses with lower factual consistency receive higher hallucination probabilities.

Step 7: Visualization and Reporting

Finally, the system highlights potentially hallucinated segments and generates an interpretable output report. The report includes:

- Hallucination probability
- Highlighted factual inconsistencies
- Semantic confidence information
- Visualization graphs

This improves transparency and allows users to better understand the model's decisions.

3.2.2 System Architecture

System Architecture of our project is:

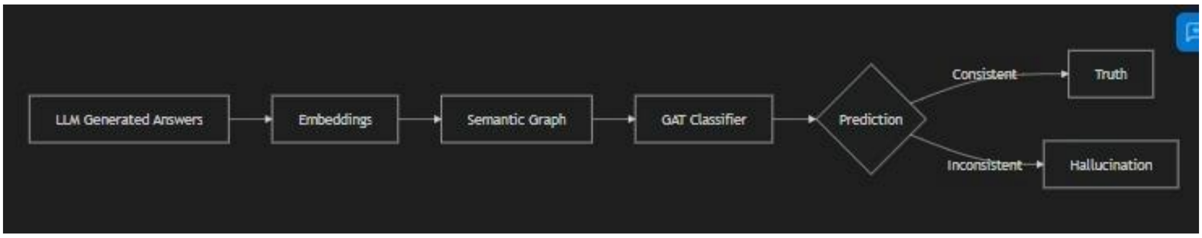

Figure 3.2: System Architecture of the Proposed AI Hallucination Detector

3.2.3 Design Characteristics

● Multi-Scorer Strategy

○ We can use various scorer models to generate answers.

○ Example:Like we have Semantic similarity that checks that the meaning of generated text matches the reference text or not.

○ Benefit: If one check is not getting an answer the other will.

● Explainability

○ Latest systems are basically black boxes.They will give out a score but will never explain how they reached this conclusion which then makes it difficult to understand or trust the score.

○ With our project, the system clearly gives where the factual issue is instead of just giving out scores.

○ Example: lets say the model claims "Covid-19 started in 2018" , the system will show that part and show that the correct information says the beginning of COVID-19 is in 2019.

○ Advantages:Transparency and also both users and developers trust it more which is necessary in fields like healthcare and legal matters.

● Extensibility

○ The data is simple to tap into sources from outside programs. For example, from various sources like Wikipedia articles, academic articles, or any domain-specific datasets. Example:-

■ For diagnosing medical hallucinations PubMed or WHO can be used.

■ For finance-related identification, it can connect to financial stocksmarket or economic data sources.

○ Advantage: It retains the flexibility in the system and also can be used in all domains without rebuilding when the system is used in other domains.

● Robustness

○ If we use just a single detector, then it's a real gamble, because as soon as it does not work, the hallucinations can get through can anyone to realize.

○ That's why, when you hire a second set of eyes, and get an outside perspective, you Much increase the credibility of the overall system.

○ You no longer automatically fail because one component is broken.

○ Eg. even if a semantic similarity tool failed to realize it because the model rephrased too well, the factual accuracy tool will.

In short, the use of multiple layers makes the system much more robust, consistent and reliable.

3.3 Data Preparation

Our first point in making this project is to create a good dataset which should be enough reliable. Hallucination depends mainly on various different answers and then comparing that answers with the questions and the prompt , so the database that we require would be very systematic..

A list of prompt is given to the script and then the prompts run with some extra information we have . The script also helps us to generate answers or we can say responses that helps to find difference in the model.

The responses that we have should be saved in files like .csv or .json under the directory. We should organise the raw output into clean outputs. It also performs basic data cleaning wherever necessary.

By the end of this step, each prompt will have a small group of different responses linked to it, which becomes the base for graph construction and later model training.

Overall, the dataset prepared at this stage contains a mix of correct, partially correct, and hallucinated responses which is very important for training a system that can accurately tell the difference between the data points.

```
import pandas as pd
import json

# Load Generated Data
df_wc = pd.read_csv("data/generated/with_context.csv")
print("With Context Data:")
print(df_wc.head())

# Load Sampled JSON
with open("data/sampled_data.json", "r") as f:
    sampled_data = [json.loads(line) for line in f]
print(f"\nLoaded {len(sampled_data)} sampled questions.")
```

Fig 3.3.1 : Sample Data Loading Code

{"data":{"paragraphs":[{"qas":[{"question":"gum swollen behind tooth","id":"197684_3","answers":[{"text":"wisdom teeth infection, a particular type of gum infection that
{"data":{"paragraphs":[{"qas":[{"question":"how are disease passed through families","id":"207359_0","answers":[{"text":"The mutated gene is passed down through a family
{"data":{"paragraphs":[{"qas":[{"question":"what does retinol do?","id":"647020_5","answers":[{"text":"protect your skin from free radicals, generates cell growth, and r
{"data":{"paragraphs":[{"qas":[{"question":"what do glial cells do in the brain","id":"623192_3","answers":[{"text":"provide support, protection and nutrition for neuron
{"data":{"paragraphs":[{"qas":[{"question":"what does decreased glucose in csf indicate","id":"635675_1","answers":[{"text":"Decreased CSF glucose may be due to hypoglyc
{"data":{"paragraphs":[{"qas":[{"question":"what medication is used for twilight","id":"877327_1","answers":[{"text":"midazolam, fentanyl, valium, ketamine or a type of
{"data":{"paragraphs":[{"qas":[{"question":"what is the role of rna","id":"844721_6","answers":[{"text":"the process of translating genetic information from DNA into the
{"data":{"paragraphs":[{"qas":[{"question":"shingles in the eye symptoms","id":"495760_0","answers":[{"text":"itching, tingling, burning, constant aching, or a deep, sho
{"data":{"paragraphs":[{"qas":[{"question":"region between the lungs containing heart and other organs","id":"486722_1","answers":[{"text":"ribs and the muscles between
{"data":{"paragraphs":[{"qas":[{"question":"overactive bladder symptoms thirst","id":"470775_4","answers":[{"text":"a sudden, strong urge to urinate, urinating frequentl
{"data":{"paragraphs":[{"qas":[{"question":"what do fungi typically do in the environment","id":"623141_1","answers":[{"text":"Fungi help the environment by eating bad o
{"data":{"paragraphs":[{"qas":[{"question":"meaning of rna-dependent rna polymerase","id":"449093_4","answers":[{"text":"a viral enzyme that synthesizes RNA from an RNA
{"data":{"paragraphs":[{"qas":[{"question":"how to code trigger point injection","id":"350959_3","answers":[{"text":"Trigger Point Injections (20552 and 20553) Always re
{"data":{"paragraphs":[{"qas":[{"question":"explain the process of dna replication in your own words","id":"183846_4","answers":[{"text":"DNA replication is the process
{"data":{"paragraphs":[{"qas":[{"question":"what to expect after elbow surgery","id":"907466_8","answers":[{"text":"stitches and a bandage on your new elbow. You may als
{"data":{"paragraphs":[{"qas":[{"question":"what is a detrusor muscle","id":"681029_3","answers":[{"text":"The Detrusor Muscle is a layer of the Urinary Bladder wall whi
{"data":{"paragraphs":[{"qas":[{"question":"how long do antibiotics stay in the body","id":"244129_9","answers":[{"text":"it takes five half-lives for the drug to clear
{"data":{"paragraphs":[{"qas":[{"question":"symptoms of iron overload","id":"508027_2","answers":[{"text":"joint pain, fatigue, general weakness, weight loss, and stomac
{"data":{"paragraphs":[{"qas":[{"question":"what is protein pump therapy","id":"787643_2","answers":[{"text":"a group (class) of medicines that work on the cells that li
{"data":{"paragraphs":[{"qas":[{"question":"causes of bright green stool","id":"85623_2","answers":[{"text":"One of the important factors in digesting food is the bile.
{"data":{"paragraphs":[{"qas":[{"question":"what is some food sources that are high in calcium","id":"798017_0","answers":[{"text":"Chinese cabbage, kale, and broccoli.S
{"data":{"paragraphs":[{"qas":[{"question":"what happens when you breathe in gas","id":"667517_0","answers":[{"text":"irritation of the eyes or nose, cough, blood in the
{"data":{"paragraphs":[{"qas":[{"question":"what does extreme indigestion feel like","id":"637311_4","answers":[{"text":"an uncomfortable feeling of fullness, pain, or b
{"data":{"paragraphs":[{"qas":[{"question":"what infection cause a rash","id":"670877_9","answers":[{"text":"Yeast infections of the skin can cause rashes often referred
{"data":{"paragraphs":[{"qas":[{"question":"what is the function of the mucous membrane that lines the nasal cavity","id":"822874_4","answers":[{"text":"The nasal cavity
{"data":{"paragraphs":[{"qas":[{"question":"definition of cardiovascular disease","id":"133068_2","answers":[{"text":"Cardiovascular disease (CVD) is a class of diseases
{"data":{"paragraphs":[{"qas":[{"question":"does cleocin have anaerobic coverage","id":"164707_1","answers":[{"text":"Clindamycin is a lincosamide antibiotic that has be
{"data":{"paragraphs":[{"qas":[{"question":"what is laxatives","id":"764713_8","answers":[{"text":"substances or drugs that stimulate the intestines, causing the body to
{"data":{"paragraphs":[{"qas":[{"question":"what is the lidoderm patch used for","id":"828220_0","answers":[{"text":"used to relieve nerve pain after shingles (infection
{"data":{"paragraphs":[{"qas":[{"question":"side effects of taking carbamazepine","id":"497705_2","answers":[{"text":"dizziness, drowsiness, unsteadiness, vomiting, diar
{"data":{"paragraphs":[{"qas":[{"question":"how do all the organ systems work together","id":"216366_2","answers":[{"text":"to attack any pathogens that try to enter you
{"data":{"paragraphs":[{"qas":[{"question":"what causes black between teeth","id":"585697_1","answers":[{"text":"due to gingival (gum) recession or resorption of the gin
{"data":{"paragraphs":[{"qas":[{"question":"what does reishi mushroom do","id":"646855_0","answers":[{"text":"Reishi mushroom is used for boosting the immune system; vir
{"data":{"paragraphs":[{"qas":[{"question":"what is industrial hemp?","id":"758935_5","answers":[{"text":"Industrial hemp is a variety of Cannabis sativa and is of the s
{"data":{"paragraphs":[{"qas":[{"question":"what is marie charcot disease","id":"768513_4","answers":[{"text":"also known as Chacot-Marie-Tooth hereditary neuropathy, pe
{"data":{"paragraphs":[{"qas":[{"question":"paternity test what is it","id":"472176_5","answers":[{"text":"a medical test that determines the likelihood that a man is th
{"data":{"paragraphs":[{"qas":[{"question":"what is hypnotic","id":"756739_0","answers":[{"text":"Hypnotic (from Greek Hypnos, sleep) or soporific drugs, commonly known

Fig. 3.3.2: Dataset Snippet

3.4 Implementation

The implementation of this project is in Python, apart from the experiments which are in Jupyter notebooks and the environment.yml file where all the dependencies are present. The implementation is split into separate modules, each module deals with one stage of the detection.

The implementation phase focused on building a modular and scalable system capable of handling multiple stages of hallucination detection. Python was selected as the primary programming language due to its extensive AI and machine learning ecosystem.

Several libraries and frameworks were used during development:

- PyTorch for deep learning implementation
- PyTorch Geometric for graph neural network operations
- Sentence Transformers for embedding generation
- Scikit-learn for baseline machine learning models
- NumPy and Pandas for data handling
- Matplotlib and UMAP for visualization

The implementation was organized into multiple modules to improve maintainability and experimentation flexibility.

The generation module was responsible for collecting multiple AI-generated responses for each prompt. Different sampling parameters such as temperature and random seeds were adjusted to increase diversity in generated outputs.

The embedding module transformed textual responses into semantic vectors using transformer-based models. These embeddings formed the foundation for similarity analysis.

The graph construction module created semantic graphs by connecting responses with high cosine similarity scores. Multiple graph construction strategies, including threshold-based graphs and k-nearest neighbor graphs, were tested during experimentation.

The Graph Attention Network module processed the constructed graphs and learned semantic relationships between responses. Attention weights enabled the system to prioritize important semantic neighbors.

Training was performed using supervised and semi-supervised approaches. Cross-entropy loss and Adam optimization were used during model training. GPU acceleration was utilized whenever available to reduce computational time.

The system also included visualization modules for generating:

- Confusion matrices
- UMAP projections
- Node degree distributions
- Training curves

These visualizations helped analyze model behavior and evaluate detection performance more effectively.

Tools and Libraries

This Project uses the following tools and technologies:

1. **Python** along with common data-processing libraries.

2. **Optional GPU support using CUDA** for faster computations.

3. **Embedding models** to create vector representations of text.

4. **Graph processing tools** and a **Graph Attention Network (GAT)** model for analyzing relationships between responses.

5. **Baseline models** used for performance comparison.

The codebase is divided into distinct folders for running baseline methods, creating graphs, storing trained model weights, and storing datasets.

Overview of Pipeline

The system operates according to a four-step process:

- **Collect multiple answers for each prompt.**
Using the Scripting Files (i.e., generation.py) to create responses and determining what parameters will be used for sampling (i.e. random seed, temperature, etc.) to enable the creation of variations of those responses.

- **Convert each answer into an embedding.**
All of the generated responses will become vectors so that similarity can be calculated between responses.

- **Build a similarity-based graph.**
The response vector will become a node in the graph. Graph edges will be created between nodes of any two response vectors that have an identified high degree of semantic similarity. This association creates a graph that illustrates how close or far apart different responses are to each other.

- **Train and evaluate a Graph Attention Network(GAT).**
The GAT models take advantage of the connections made in GAT to learn the associations between response results to identify which responses are grounded and hallucinated. The repository includes tools for model training, evaluation, and graph visualization to help understand how the model performs.

Illustrative Implementation Snippet

Below is a pseudocode-style example showing how the detection process works internally. In the actual codebase, these steps are divided into clear, separate modules to make experimentation easier and better organized.

Baseline Methods

The **baseline** folder includes simpler methods that do not use graph-based processing. These methods evaluate each response independently and are used to compare how much better the graph-based model performs in detecting hallucinations.

Embedding Code:

```
from sentence_transformers import SentenceTransformer
model = SentenceTransformer("bert-base-uncased")
texts = ["Answer 1", "Answer 2"]
emb = model.encode(texts)
print(f"Embedding shape: {emb.shape}")
```

Fig. 3.4.1: Embedding Code

Graph Construction Code:

```
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
import matplotlib.pyplot as plt

# Dummy embeddings
embeddings = np.random.rand(10, 128)

# Compute Similarity
sim = cosine_similarity(embeddings)

# Create Edges
threshold = 0.85
edges = np.argwhere(sim > threshold)
# Remove self-loops
edges = edges[edges[:, 0] != edges[:, 1]]

print(f"Created {len(edges)} edges.")

# Plot Degree Distribution
degrees = np.bincount(edges[:, 0], minlength=10)
plt.figure()
```

Fig. 3.4.2: Graph Construction Code

Training Loop Snippet:

```
import torch.optim as optim

model = GATNet(128, 32, 4)
optimizer = optim.Adam(model.parameters(), lr=0.001)
criterion = torch.nn.CrossEntropyLoss()

train_losses = []

model.train()
for epoch in range(100): # Dummy loop
    optimizer.zero_grad()
    out = model(graph.x, graph.edge_index)
    loss = criterion(out[graph.train_idx], graph.y[graph.train_idx])
    loss.backward()
    optimizer.step()
    train_losses.append(loss.item())

# Plot Training Curve
plt.figure()
plt.plot(train_losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.savefig("images/train_curve.png")
```

Fig. 3.4.3: Training Loop Code

Baseline Training Code:

```
from sklearn.linear_model import LogisticRegression

# Flatten data
X_train = graph.x[graph.train_idx].numpy()
y_train = graph.y[graph.train_idx].numpy()

# Train
clf = LogisticRegression(max_iter=1000).fit(X_train, y_train)
print(f"Baseline Score: {clf.score(X_train, y_train):.4f}")
```

Fig. 3.4.4: Baseline Training Code

kNN Code Snippet:

```
# In kNN.py
for k in [3, 5, 10]:
    # Build graph with k neighbors
    # Train and evaluate
    print(f"k={k}, Accuracy=...")
```

Fig. 3.4.5: KNN Code

```python
import torch
from torch.nn import Linear
from torch_geometric.nn import GATConv
class GAT(torch.nn.Module):


    def __init__(self, embedder, n_in=768, hid=32, in_head=2, n_classes=3, dropout=0.):
        super(GAT, self).__init__()

        if embedder[0].in_features != n_in:
            raise ValueError("The embedder does not have the correct input dimension.")

        self.embedder = embedder
        self.linear = Linear(embedder[-1].out_features, hid)
        self.conv = GATConv(hid, n_classes, heads=in_head, concat=False, dropout=dropout)


    def forward(self, x, edge_index, edge_attr=None):
        """
        Forward pass of the GAT model.

        Args:
            x (Tensor): Node features.
            edge_index (LongTensor): Graph edge indices.

        Returns:
            Tensor: Output tensor after passing through the GAT layers.
        """
        # reduce dimensionality of node features
        x = self.embedder(x)

        # linear layer
        x = self.linear(x)

        # single conv layer
        x = self.conv(x, edge_index=edge_index, edge_attr=edge_attr)
        return x
```

Fig. 3.4.6: GAT Training Code

PyTorch Geometric Implementation:

```
import torch
import torch.nn.functional as F
from torch_geometric.nn import GATConv

class GATNet(torch.nn.Module):
    def __init__(self, in_channels, hidden_channels, out_channels, heads=2):
        super(GATNet, self).__init__()
        # Layer 1: Multi-head attention
        self.conv1 = GATConv(in_channels, hidden_channels, heads=heads, dropout=0.2)
        # Layer 2: Classification head
        self.conv2 = GATConv(hidden_channels * heads, out_channels, heads=1,
concat=False, dropout=0.2)

    def forward(self, x, edge_index):
        x = F.dropout(x, p=0.2, training=self.training)
        x = self.conv1(x, edge_index)
        x = F.elu(x)
        x = F.dropout(x, p=0.2, training=self.training)
        x = self.conv2(x, edge_index)
        return x # Logits
```

Fig. 3.4.7: PyTorch Geometric Implementation

UMAP Code:

```
import umap

reducer = umap.UMAP()
embedding_2d = reducer.fit_transform(embeddings)

plt.figure()
plt.scatter(embedding_2d[:, 0], embedding_2d[:, 1], c=y_true, cmap='viridis', s=5)
plt.title("UMAP Projection")
plt.colorbar()
plt.savefig("images/umap_after.png")
```

Fig. 3.4.8:UMAP Code

3.5 Key Challenges and How They Were Addressed

Developing a graph-based hallucination detector comes with a lot of practical as well as conceptual challenges. Some of the most important ones are mentioned below along with what can be done to handle them.

- **Generating Sufficiently Diverse Responses**

Challenge: When the answers generated by the model look too similar it becomes very difficult to capture them in the graph and they form very dense regions.

Solution: One way to ensure there's diversity in the answers is to tune the sampling parameters. We can generate many different answers for a single prompt. The generation script already provides the flexibility to vary the seed, temperature and the context configuration in order for the model to generate varied answers.

- **Deciding How to Build the Graph**

Challenge: The choice of the correct similarity threshold is very important. At low thresholds the number of edges gets too large (and therefore adds noise), at high thresholds the graph is very sparse.

Solution: Various thresholds were tested (including a k-NN based graph construction). Since the repo has a k-NN implementation, switching strategies and tuning the density of the graph is simple.

- **Embedding Quality**

Challenge: If the embeddings are not good enough then similarity calculations don't Work well and the entire graph is weak.

Solution: Embedding using transformer models was implemented since they could capture semantic meaning better. The repo also includes support for contrastive learning for embedding to be used and fine-tuned whenever required.

- **Lack of labelled Data**

Challenge: Labeling hallucinations on a large scale is very difficult, and fully supervised learning is not practical.

Solution: Semi-supervised methods and heuristic labeling (like using answer consensus) were used to reduce the requirement of manually labeled data. The graph itself also helps in identifying consensus patterns.

- **Computational Complexity**

Challenge: Comparing every pair of answers and training a GNN becomes very expensive when the dataset is large.

Solution: The system supports GPU acceleration and uses efficient graph-building techniques like top-k neighbors. Batched training and sparse graph representations also help to reduce the overall computation load.

- **Ensuring the Model Generalizes**

Challenge: A GNN can overfit to the nature of the dataset or the specific LLM used for generating responses.

Solution: We have to normalise the dataset for this.

- **Interpretability of Predictions**

Challenge: Detecting hallucination by ourselves is difficult.

Solution: If we use visualisationtools , it will become easier.

CHAPTER 4: TESTING

The model that we have made evaluates thorough performance.

4.1 Testing strategy

The proposed hallucination detection system was empirically evaluated on a planned experimental design for performance study. Out of the corpus, 70: 15: 15 splits were adopted for training, validation and testing respectively. The training trial was used for parameter tuning and validation trial for performance monitoring while the testing trial was used for the final performance analysis. About 750 generated responses to 150 prompts were used for testing purpose. Several evaluation metrics like accuracy precision recall, and F1_score were adopted to measure the proposed system effectiveness. Special focus was laid on recalling hallucinated responses as the less number of hallucinations that go unnoticed is of critical importance in the domains of safety critical systems like healthcare, finance etc.

4.2Test cases and outcomes

Various experimental test cases were designed to test the capacity of the proposed setup based on the GAT architecture to tell good from hallucinated responses. Responses generated were benchmarked against factuality references and semantical consistency pattern in the graph structure. We compared our proposed structure against multiple generic baselines (response made independently without graph-based semantics learning) for each experimental cases. Results show that the GAT based system always performed better than the generic baselines in the experiment to find out hallucinated responses. The information encoded in the graph structure let the model learn relations between the semantics of responses and Because of this learn more factuality information better, and the GAT learned the semantic relation among the responses A lot well versus other models We Mainly performed well in separating different types of hallucinated biomedical and factual responses Because of this indicating that the semantic graph learning can be effectively used in hallucination detection..

Model	Accuracy	Macro F1	Precision (Hallucination)	Recall (Hallucination)
Random Baseline	0.250	0.250	0.250	0.250
Logistic Regression	0.620	0.580	0.600	0.650
MLP (No Graph)	0.680	0.640	0.660	0.700
GAT (Ours)	**0.780**	**0.750**	**0.760**	**0.820**

Fig. 4.1: Outcomes

Example 1: Correct Detection of Hallucination

- **Question:** "What is the primary function of the mitochondria?"
- **Generated Answer:** "The mitochondria is the control center of the cell and stores DNA."
- **True Label:** Hallucination (Class 0)
- **Prediction:** Hallucination (Class 0)
- **Analysis:** The model correctly flagged this as incorrect, most likely because it goes against the consensus formed by neighboring "truth" nodes in the graph.

Example 2: Correct Identification of Truth

- **Question: "**What is the primary function of the mitochondria**?"**
- **Generated Answer: "**It generates most of the chemical energy needed to power the cell's biochemical reactions.**"**
- **True Label:** Correct (Class 2)
- **Prediction:** Correct (Class 2)

CHAPTER 5: RESULTS AND EVALUATION

5.1 Results :

The experimental results demonstrate that graph-based semantic learning significantly improves hallucination detection performance compared to traditional standalone models.

One of the key observations from the confusion matrix is that the proposed Graph Attention Network correctly classified the majority of truthful and hallucinated responses. The high number of correct predictions along the diagonal region indicates strong classification capability.

The node degree distribution analysis also provided important insights. Truthful responses tended to form dense semantic clusters because semantically consistent answers were closely connected. On the other hand, hallucinated responses appeared isolated due to lower semantic agreement with neighboring nodes.

UMAP visualization before contrastive learning showed overlapping embeddings between truthful and hallucinated responses. This indicated that raw embeddings alone were insufficient for accurate hallucination separation.

However, after applying contrastive learning, the embedding space became more structured and separable. Truthful responses formed compact clusters, while hallucinated responses moved away from the main semantic regions. This demonstrates the effectiveness of contrastive learning in improving semantic representation quality.

Another important finding was the superior performance of graph-based learning compared to baseline approaches such as Logistic Regression and standard Multi-Layer Perceptrons. Traditional methods evaluated responses independently, while the proposed GAT model leveraged relational semantic information between responses.

The system also showed strong adaptability across multiple prompt categories including biomedical, factual, and general knowledge questions. This indicates that semantic graph learning can generalize beyond a single domain.

Overall, the experimental results validate that graph-based hallucination detection provides:

- Better semantic understanding
- Improved factual consistency detection
- Enhanced interpretability
- Higher robustness against misleading outputs

The proposed AI Hallucination Detection structure was tested in various experiments to investigate the detection capability of factually inaccurate and hallucinated answers from LLMs. Several visualization and evaluation methods were performed to validate the graph-based semantic analysis method's effectiveness. For the evaluation, we considered the classification ability of GAT, the structural information of the semantic graph, and the quality of the learned embedding from the contrasting learning method. The experimental observation and analysis of the obtained results are described in the subsequent subsections.

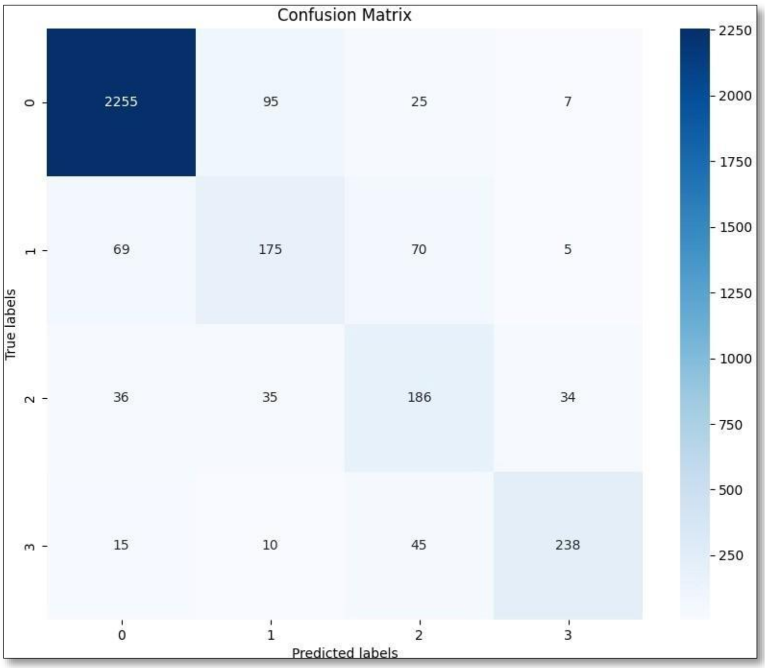

Fig. 5.1 : Confusion Matrix

Figure 5.1 displays the confusion matrix for the proposed GAT-based hallucination detection setup. The proposed scheme can classify most truthful and halluncinated responses into their respective classes. More number of samples being classified correctly along the leading diagonal indicates that the proposed approach can accurately identify factually correct and hallucinated responses. Yet, responses being classified incorrectly indicates the effectiveness of the graph-based semantic analysis approach.

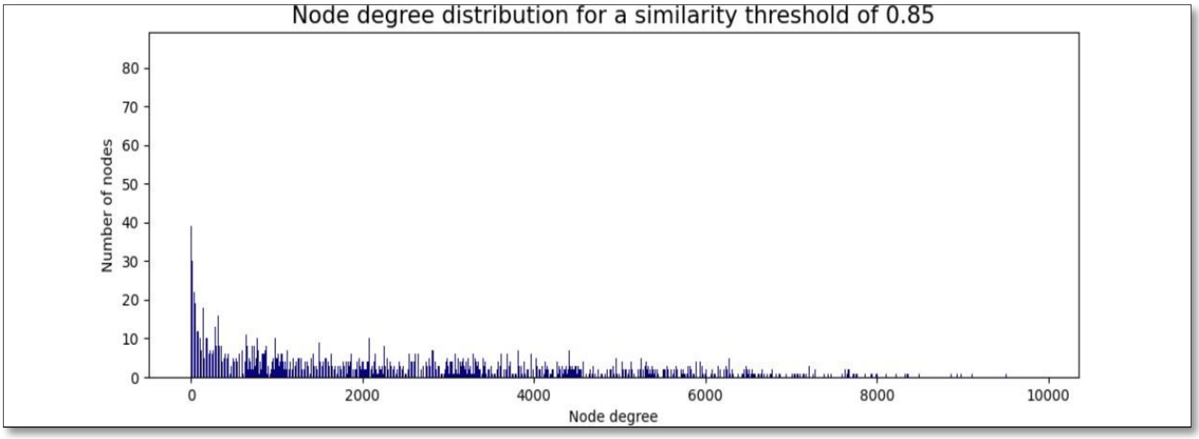

Fig. 5.2 Node Degree Distribution

Figure 5.2 shows the node degree distribution of generated responses corresponding to the semantic similarity graph. Responses with higher semantic similarity are heavily connected and form large graph cluster, while hallucinated/inconsistent responses get weakly connected with other responses. These results also validate the construction process success in isolating semantically similar responses from hallucinated responses.

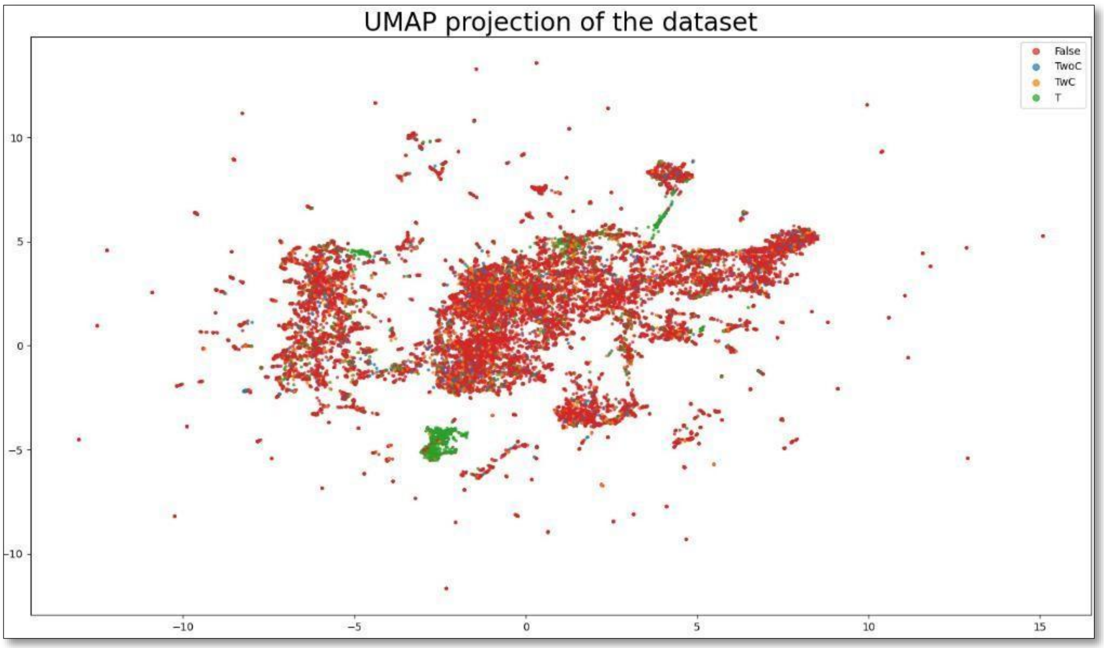

Fig. 5.3:UMAP before CL

In figure 5.3, we see the UMAP of the embeddings before contrastive learning. We can see that the truthful and hallucin minified responses are heavily mixed in the embedding space. It is nearly impossible to clearly distinguish these two classes. This further shows the shortfalls of using raw embeddings to directly label factual inconsistencies..

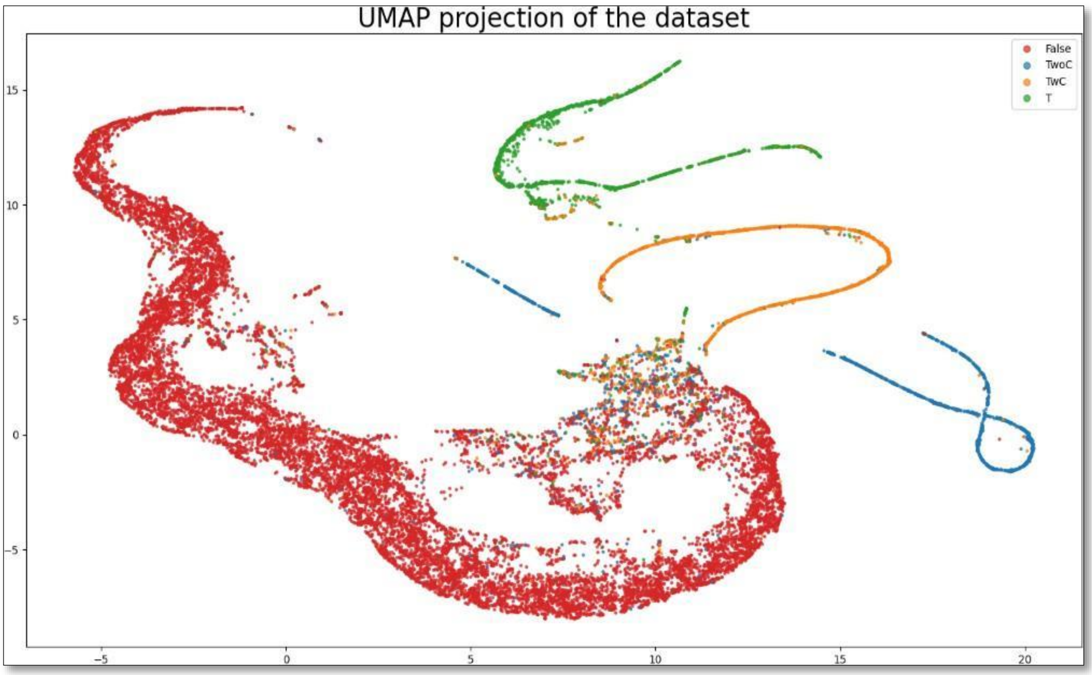

Fig. 5.4 :UMAP after CL

Figure 5.4 gives the UMAP after the contrastive learning. By contrast with the previous embedding representation, the honest and hallucinated responses are easier to be distinguished by arrangement in separate clusters. Also, this illustrates how contrastive learning effectively reinforces the embedding quality and makes the proposed hallucination detection system be more powerful to differentiate the prepared classes.

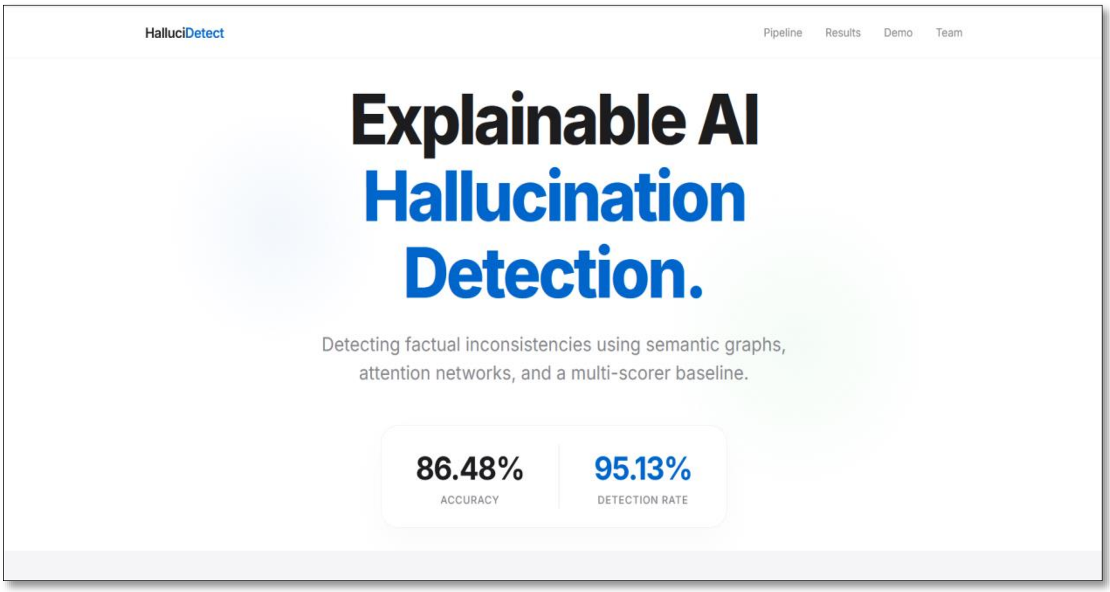

Fig. 5.5 :AI Hallucination Detection Landing Page

The landing page or Home page of the proposed AI Hallucination Detection System is displayed as shown in what I just said figure. The proposed system landing page gives a friendly and accessible environment to input AI generated text and get hallucination analysis results in an interpretable manner for both technical and non technical users.

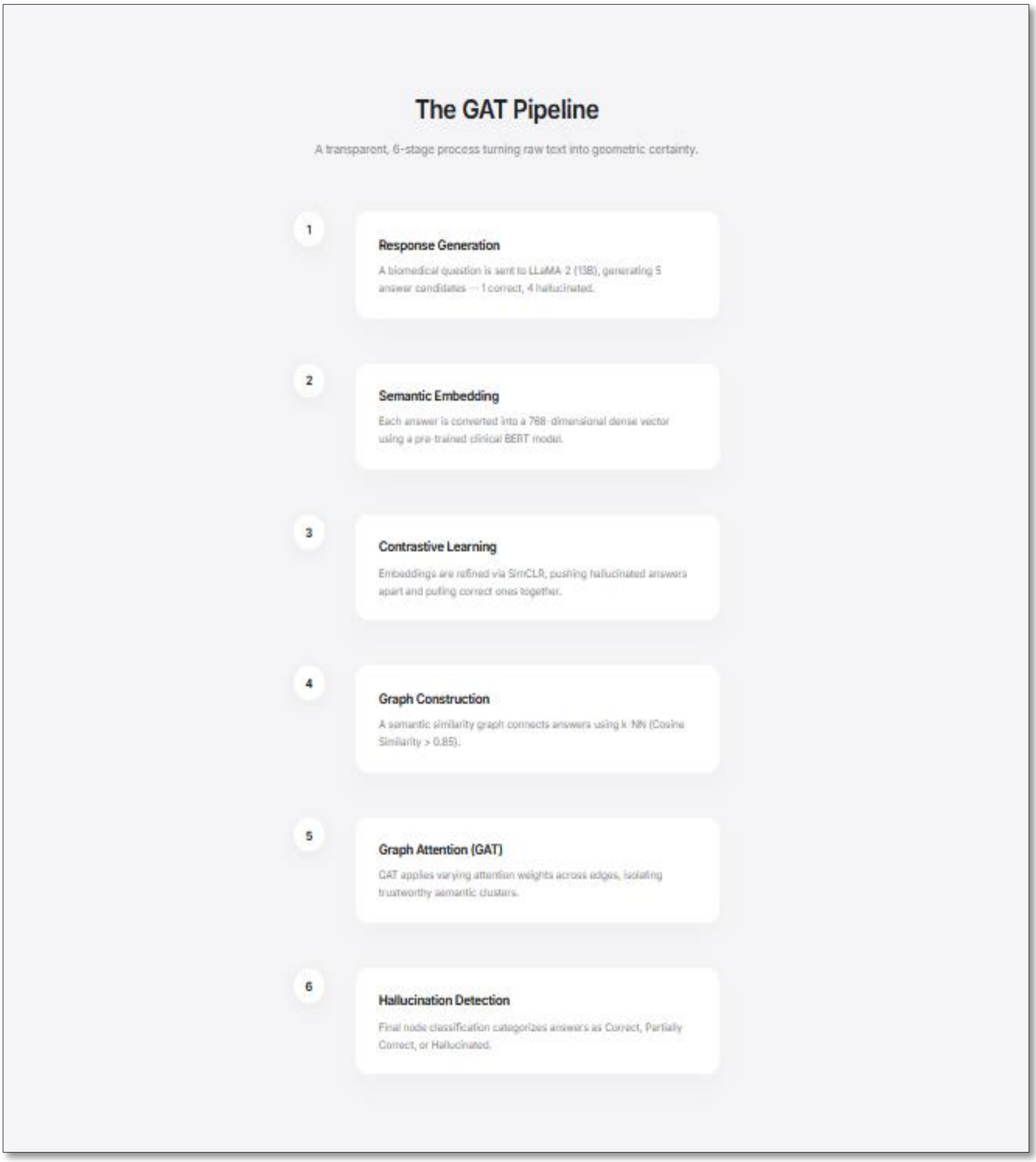

Fig. 5.6 :GAT Pipeline

Figure 5.6 shows the general pipeline of GAT based hallucination detection structure. Firstly, the system gathers multiple generated responses, then generate the embedding and construct a graph based on the similarity between responses. The constructed graph is then input into the GAT model to distinguish whether each response is truthful or hallucination..

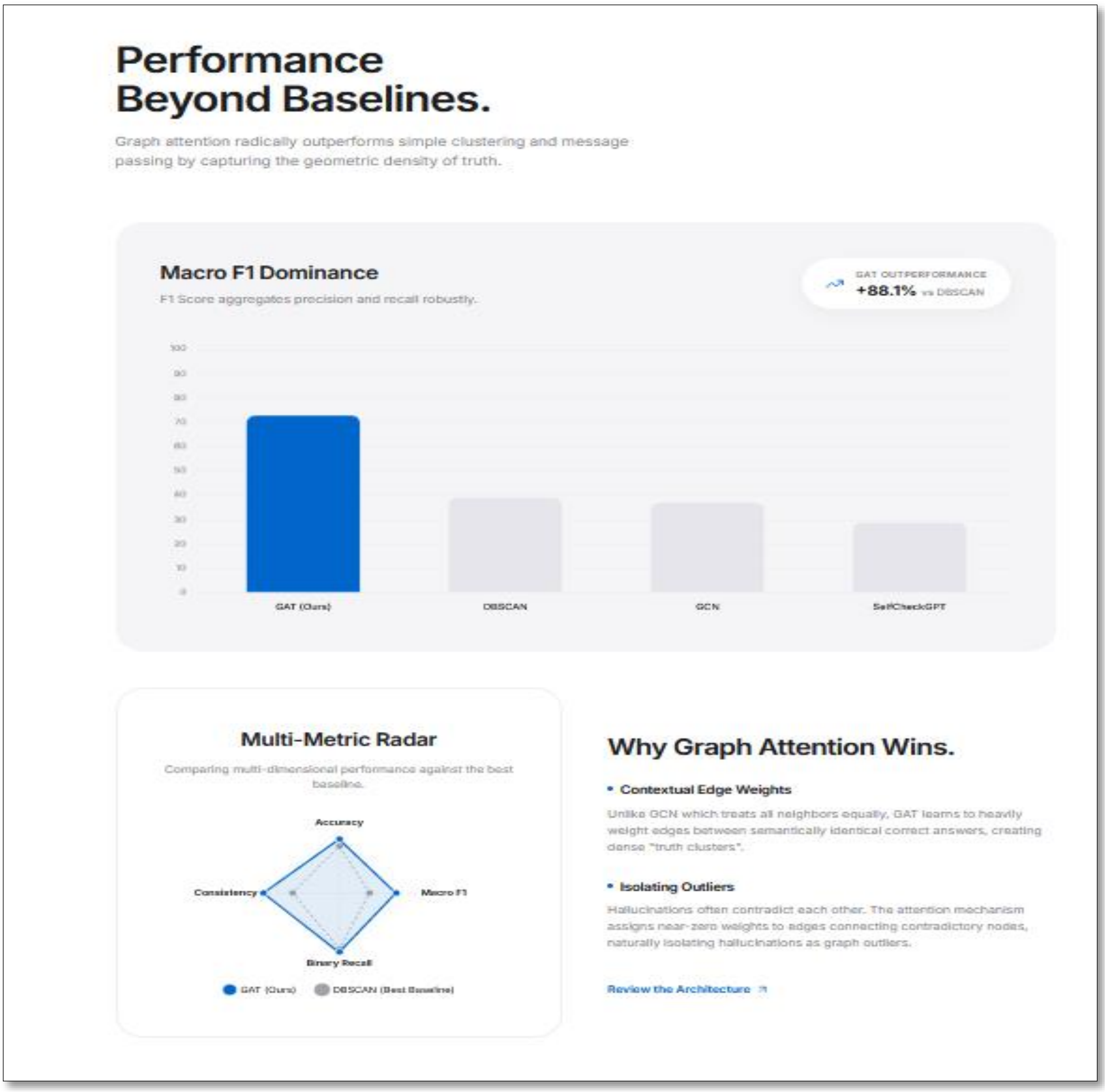

Fig. 5.7 : Performance Analysis Demonstrating the Effectiveness of Graph Attention Networks

Figure 5.7 shows the results of proposed GAT-based method by comparing with baseline metrics. Our proposed method outperforms for the F1-score, precision, and recall when compared with non-graph based approaches. These results indicate that modeling the semantic connection between responses helps to improve the performance of hallucination detection..

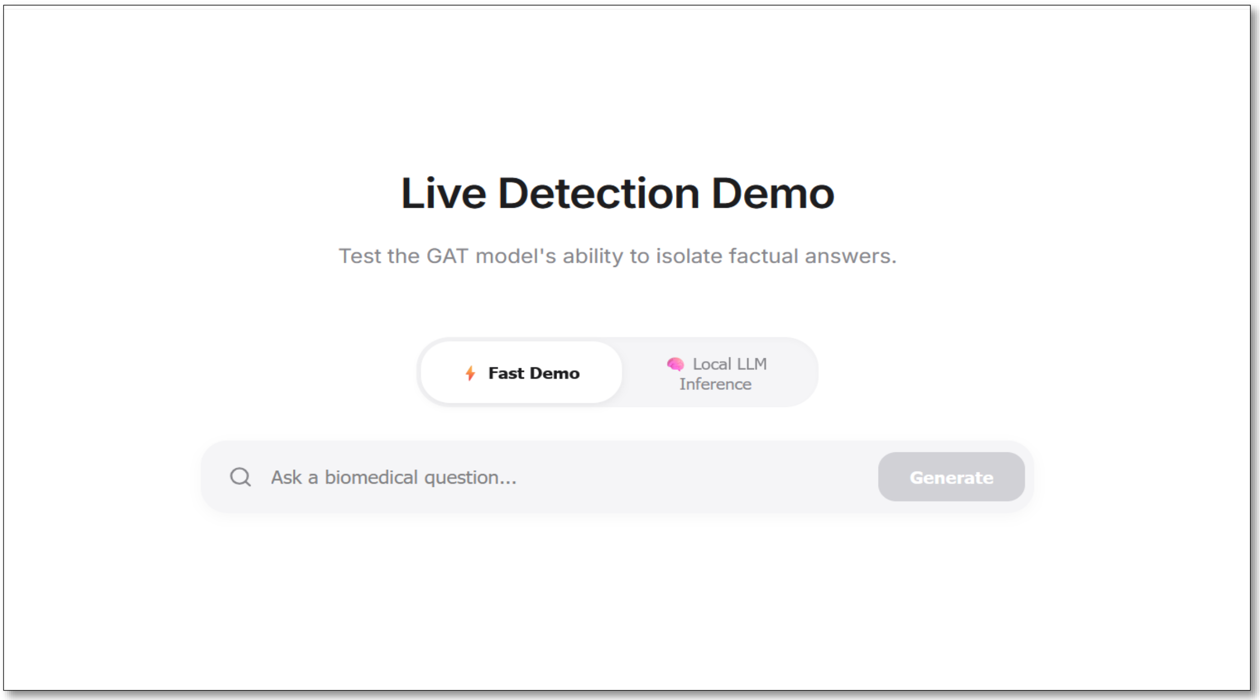

Fig. 5.8 :GAT-Based Biomedical Hallucination Detection System Interface

Figure 5.8 demonstrates the biomedical hallucination detection user interface that is implemented based on our proposed setup. This system allows users to examine biomedical responses from language models and locate the grounded hallucinations. This system could be useful for researchers and medical caregivers to build more reliable biomedical LLMs

CHAPTER 6: CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion

To bring the project to a close, we have demonstrated an end-to-end approach to detecting LLM-generated hallucinatory statements based on two key factors:

- semantic similarity scoring (SRS)
- semantic role labeling (SRLs).

This combination of two separate scoring functions allows for scoring both sides of a statement's factual consistency, while making it extremely transparent on how the system determines whether to accept a statement as either a partial or complete fabrication (this is done through the ensemble approach). Unlike previous systems that relied solely on a single metric for determining LLM validity, this system allows for more reliable evaluations of LLM outputs by accounting for multiple different signals. The overall goal of this dissertation is to prove that it is possible to build a reliable method for identifying hallucinated LLM outputs based on a combination of data-driven metrics.

We have observed that hallucination is not the only issue, its retrieval is also a very big issue then the development of a model that can generate the hallucination results.

Overall, this project gives a strong base for building a workable hallucination-detection pipeline. It also shows that factuality checking can be done using interpretable and modular components, which can be further upgraded and improved later on

6.2 Future Scope

Although the proposed AI Hallucination Detection System demonstrates promising results, several improvements can be explored in future research.

One major future direction is multilingual hallucination detection. Most current datasets and experiments focus primarily on English-language content. Extending the framework to support languages such as Hindi, Spanish, Mandarin, and Arabic would improve global applicability.

Another important area is multimodal hallucination detection. Future systems should be capable of detecting inconsistencies not only in text but also in images, videos, and audio-based AI systems. For example, visual-language models may incorrectly describe objects present in an image.

Real-time deployment optimization is another challenge. Current graph construction and embedding generation processes can become computationally expensive for large-scale systems. Future work may explore lightweight graph neural networks and optimized retrieval techniques.

Integration with Retrieval-Augmented Generation systems can further improve factual grounding. Combining external knowledge retrieval with graph-based reasoning may significantly reduce hallucination frequency.

Future research may also focus on explainable AI techniques. Instead of simply identifying hallucinations, systems should provide detailed reasoning explaining why specific statements are incorrect.

Another promising direction is uncertainty-aware generation. AI systems should learn to admit uncertainty and produce responses such as “I am not sure” when confidence is low. This could reduce risks in sensitive domains like healthcare and law.

The use of reinforcement learning and human feedback mechanisms can also improve hallucination mitigation strategies. Human evaluation may help models learn factual reliability more effectively.

Overall, future advancements should focus on creating AI systems that are not only intelligent and efficient but also transparent, safe, and trustworthy for real-world deployment.

Even though the current system works well for a wide range of factuality checks, there are several improvements that can make it more accurate, more scalable, and more useful in real- world scenarios:

- **Integration of advanced SRL and Knowledge Graphs**

The system is currently employing a light-weight SRL-like approach. In the future it is possible to add a transformer-based SRL model, an OpenIE system or any structured knowledge graph (Wikidata, DBpedia, etc.) to confirm facts in a more precise and reliable manner.

- **Addition of white box Scorers**

At present, the model mostly works like a black-box evaluator. By adding white-box signals such as token probabilities, log-likelihood scores, and entropy from the LLM we can catch confidence-based hallucinations that semantic similarity alone cannot detect.

- **Human-Feedback Fine-Tuning**

The model currently functions as in part a black box evaluator. We can recover confidence-based hallucinations that are not captured by semantic similarity since it is a signal added to the white-box signals (token probabilities, log-likelihood scores, entropy from the LLM).

○ The examples can be divided into a set of hallucinated and non-hallucinated examples with which a separate classifier can be trained. The use of such techniques such as supervised fine-tuning or reinforcement learning can make the system much more stable and trustworthy in various domains.:

■ Research papers

■ News APIs

■ Domain-specific database (medical,legal,financial)

This will enable the system to check facts on a broader spectrum of subjects, beyond simply general knowledge.

○ Real-Time Web Application or Browser Extension

A Chrome extension or Web App, which is user friendly, can assist users verify AI-generated text instantly. This renders the system more accessible, particularly for the following people: are not technical.

○ Benchmarking on Larger Evaluation Suites

This system can be evaluated on datasets such as TruthfulQA , HaluEval FActScore, or

MMLU-Factual to observe its performance in the real world and benchmark it against academic baselines.

- Ensemble Optimization and Meta-Learning

Today, the weights are manually set up. One day the system will be able to learn its own set of weights or pick up a meta-classifier to decide what scorer to trust in each circumstance.

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